

Automation Testing Practice – Code Report

Github Link: <https://github.com/Saiteja-231/Capstone-Project>

Project Name

Automation Testing Practice

Objective

To automate the functional testing of the Automation Testing Practice website using Selenium WebDriver, Java, TestNG, Maven, Jenkins, Page Object Model (POM), and Extent Reports.

Summary

The Automation Testing Practice Website Automation Framework is a web automation testing project developed using Java, Selenium WebDriver, TestNG, Maven, Jenkins, Extent Reports, and GitHub, following the Page Object Model (POM) design pattern. The framework automates key web application functionalities such as Form Validation, Date Picker Handling, Radio Buttons, Checkboxes, Dropdowns, File Upload, Pagination Web Tables, Alert Handling, Window and Tab Management, Dynamic Button Operations, Mouse Hover, Drag and Drop, Broken Link Verification, and Search Functionality.

The framework incorporates reusable page classes, modular test scripts, screenshot capture capabilities, TestNG reporting, and detailed Extent Reports to ensure maintainability and effective test execution. Jenkins is integrated to support Continuous Integration (CI) and automated test execution, while GitHub is used for source code management and version control.

A total of 14 automated test cases were executed successfully, achieving a 100% pass rate with no failed or skipped test cases. The framework validates both functional and user-interface components of the application and demonstrates the effective implementation of automation testing best practices. The project showcases a complete end-to-end automation testing lifecycle, including framework development, test execution, reporting, and continuous integration, thereby improving test coverage, reducing manual effort, and ensuring application quality.

Code

/Automation Test Practice/src/test/java/Capstone Project/BaseTest.java

```
package Capstone_Project;
```

```
import org.testng.annotations.*;
import POM.POM_Implementation;
import java.io.*;
import org.apache.poi.xssf.usermodel.*;
import org.openqa.selenium.*;
```

```

import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.WebDriverWait;

public class BaseTest {
    WebDriver driver;
    WebDriverWait wait;
    JavascriptExecutor js;
    POM_Implementation p;

    // DataProvider reads Name from Excel
    @DataProvider(name="NameExcel")
    public Object[][] nameexcel() throws IOException {
        FileInputStream fs = new FileInputStream("D:\\Wipro Training\\NameData.xlsx");
        XSSFWorkbook wb = new XSSFWorkbook(fs);
        XSSFSheet sh = wb.getSheet("sheet1");
        int rows = sh.getPhysicalNumberOfRows();
        int cols = sh.getRow(0).getLastCellNum();
        Object[][] data = new Object[rows-1][cols];
        for (int i=1; i<rows; i++)
            for (int j=0; j<cols; j++)
                data[i-1][j] = sh.getRow(i).getCell(j).toString();
        fs.close(); wb.close();
        return data;
    }

    // DataProvider for search
    @DataProvider(name="searchdata")
    public Object[][] data() {
        return new Object[][] { {"Laptop"} };
    }

    // DataProvider for paragraph text boxes
    @DataProvider(name="paragraph")
    public Object[][] paragraphbox() {
        return new Object[][] {
            {
                "The ocean is deep and wide",
                "Stars shine bright at night",
                "Rivers flow to the sea"
            }
        }
    }
}

```

```
};  
}
```

@BeforeMethod

```
public void beforeMethod() throws Exception {  
    driver = new ChromeDriver();  
    p = new POM_Implementation(driver);  
    driver.get("https://testautomationpractice.blogspot.com/");  
    Thread.sleep(3000);  
    driver.manage().window().maximize();  
}
```

@Test(dataProvider="NameExcel", priority=1)

```
public void TC01_EnterName(String fname) throws Exception {  
    p.Name_Element(fname);  
}
```

@Test(priority=2)

```
public void TC02_RadioAndCheckBox() throws Exception {  
    p.radio_checkbox();  
}
```

@Test(priority=3)

```
public void TC03_UploadFiles() throws Exception {  
    p.upload();  
}
```

@Test(priority=4)

```
public void TC04_PaginationTable() throws Exception {  
    p.page_table();  
}
```

@Test(dataProvider="paragraph", priority=5)

```
public void TC05_Forms(String p1, String p2, String p3) throws Exception {  
    p.forms(p1, p2, p3);  
}
```

@Test(dataProvider="searchdata", priority=6)

```
public void TC06_Search(String item) throws Exception {  
    p.search(item);  
}
```

```

    }

    @Test(priority=7)
    public void TC07_DynamicButton() throws Exception {
        p.dynamicbtn();
    }

    @Test(priority=8)
    public void TC08_Alerts() throws Exception {
        p.alertaccept();
    }

    @Test(priority=9)
    public void TC09_WindowHandling() throws Exception {
        p.newtab();
    }

    @Test(priority=10)
    public void TC10_MouseHover() throws Exception {
        p.movetoelement();
    }

    @Test(priority=11)
    public void TC11_DragAndDrop() throws Exception {
        p.draganddrop();
    }

    @Test(priority=12)
    public void TC12_BrokenLinks() throws Exception {
        p.broken_links();
    }

    @AfterMethod
    public void afterMethod() {
        driver.quit();
    }
}

```

/Automation Test Practice/src/test/java/Listener/Listenerclass.java
package Listener;

```

import org.testng.ITestListener;
import org.testng.ITestResult;

public class Listenerclass implements ITestListener {

    @Override
    public void onTestStart(ITestResult result) {
        System.out.println("Test Started: " + result.getName());
    }

    @Override
    public void onTestSuccess(ITestResult result) {
        System.out.println("PASSED: " + result.getName());
    }

    @Override
    public void onTestFailure(ITestResult result) {
        System.out.println("FAILED: " + result.getName());
    }

    @Override
    public void onTestSkipped(ITestResult result) {
        System.out.println("SKIPPED: " + result.getName());
    }
}

```

/Automation Test Practice/src/test/java/POM/POM Implementation.java

```

package POM;

import java.io.*;
import java.time.Duration;
import java.util.Set;
import org.apache.commons.io.FileUtils;
import org.openqa.selenium.*;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.*;

public class POM_Implementation {
    WebDriver driver;
}

```

```
JavascriptExecutor js;  
WebDriverWait wait;  
TakesScreenshot ts;
```

```
public POM_Implementation(WebDriver driver) {  
    this.driver = driver;  
}  
By fnamefield = By.id("name");  
By emailfield = By.id("email");  
By phonefield = By.id("phone");  
By addressfield = By.id("textarea");  
By radiobutton = By.id("male");  
By sundaycheckbox = By.id("sunday");  
By mondaycheckbox = By.id("monday");  
By tuesdaycheckbox = By.id("tuesday");  
By countrydropdown = By.id("country");  
By colorselect = By.id("colors");  
By Listselect = By.id("animals");  
By datepicker1 = By.id("datepicker");  
By submitbutton = By.className("submit-btn");  
By singlefileupload = By.id("singleFileInput");  
By dynamicbtn = By.name("start");  
By alertbutton = By.id("alertBtn");  
By searchbox = By.id("Wikipedia1_wikipedia-search-input");  
By searchbutton = By.xpath("//*[@id='Wikipedia1_wikipedia-search-  
form']/div/span[2]/span[2]/input");  
By searchsuggestion = By.partialLinkText("Laptop");  
By mousehover = By.xpath("//*[@id='HTML3']/div[1]/div/button");  
By source = By.id("draggable");  
By destination = By.id("droppable");  
  
public void Name_Element(String fname) throws InterruptedException {  
    js = (JavascriptExecutor) driver;  
    js.executeScript("window.scrollTo(0,100)");  
    Thread.sleep(2000);  
    wait = new WebDriverWait(driver, Duration.ofSeconds(10));  
    WebElement firstname =  
        wait.until(ExpectedConditions.elementToBeClickable(fnamefield));  
    firstname.click();  
}
```

```

        firstname.sendKeys(fname);
        System.out.println("Name entered: " + fname);
        Thread.sleep(1000);
    }

    public void radio_checkbox() throws InterruptedException {
        js = (JavascriptExecutor) driver;
        js.executeScript("window.scrollTo(0,900)");
        driver.findElement(radiobutton).click();
        Thread.sleep(1000);
        driver.findElement(sundaycheckbox).click();
        driver.findElement(mondaycheckbox).click();
        driver.findElement(tuesdaycheckbox).click();
        Select s = new Select(driver.findElement(countrydropdown));
        s.selectByVisibleText("India");
        Thread.sleep(1000);
        Select color = new Select(driver.findElement(colorselect));
        color.selectByVisibleText("White");
        driver.findElement(datepicker1).sendKeys("08/20/2026");
        driver.findElement(datepicker1).sendKeys(Keys.ENTER);
        Thread.sleep(1000);
        driver.findElement(submitbutton).click();
        System.out.println("TC02 Radio, Checkbox, Dropdowns done");
    }

    public void search(String searchitem) throws InterruptedException {
        WebElement text = driver.findElement(searchbox);
        text.click();
        Thread.sleep(2000);
        text.sendKeys(searchitem);
        Thread.sleep(3000);

        driver.findElement(searchbutton).click();
        Thread.sleep(8000);
        String parentWindow = driver.getWindowHandle();

        driver.findElement(By.partialLinkText("Laptop")).click();
        Thread.sleep(5000);

        Set<String> allWindows = driver.getWindowHandles();
    }

```

```

    for (String window : allWindows) {
        if (!window.equals(parentWindow)) {
            driver.switchTo().window(window);
        }
    }

    System.out.println("New window title: " + driver.getTitle());
    Thread.sleep(3000);
    driver.close();
    driver.switchTo().window(parentWindow);
    System.out.println("TC06 Search Completed Successfully");
}

public void alertaccept() throws InterruptedException {
    driver.findElement(alertbutton).click();
    Alert al = driver.switchTo().alert();
    Thread.sleep(1000);
    al.accept();
    System.out.println("Simple Alert Accepted");
    driver.findElement(By.id("confirmBtn")).click();
    Alert confirmAlert = driver.switchTo().alert();
    Thread.sleep(1000);
    confirmAlert.dismiss();
    System.out.println("Confirm Alert Dismissed");
    driver.findElement(By.id("promptBtn")).click();
    Alert promptAlert = driver.switchTo().alert();
    Thread.sleep(1000);
    promptAlert.accept();
    System.out.println("Prompt Alert Accepted");
}

public void draganddrop() throws InterruptedException {
    JavascriptExecutor js = (JavascriptExecutor) driver;
    js.executeScript("window.scrollTo(0,1400)");
    Thread.sleep(1000);
    WebElement drag = driver.findElement(By.id("draggable"));
    WebElement drop = driver.findElement(By.id("droppable"));
    Actions d = new Actions(driver);
    d.dragAndDrop(drag, drop).perform();
    System.out.println("Successfully Dragged and Dropped the Element");
}

```

```
Thread.sleep(1000);  
}
```

```
public void broken_links() throws InterruptedException, IOException {  
    JavascriptExecutor js = (JavascriptExecutor) driver;  
    js.executeScript("window.scrollTo(0,2700)");  
    Thread.sleep(1000);  
    driver.findElement(By.linkText("Errorcode 400")).click();  
    TakesScreenshot ts = (TakesScreenshot) driver;  
    File src = ts.getScreenshotAs(OutputType.FILE);  
    FileUtils.copyFile(src, new File("Screenshot/Error400.png"));  
    System.out.println("Screenshot saved for Error 400");  
    driver.navigate().back();  
    Thread.sleep(1000);  
    driver.findElement(By.linkText("Errorcode 404")).click();  
    System.out.println("Page Title: " + driver.getTitle());  
    driver.navigate().back();  
    Thread.sleep(1000);  
}
```

```
public void upload() throws InterruptedException {  
    js = (JavascriptExecutor) driver;  
    js.executeScript("window.scrollTo(0,1700)");  
    Thread.sleep(3000);  
  
    WebElement singlefile = driver.findElement(By.id("singleFileInput"));  
    singlefile.sendKeys("D:\\Wipro Training\\Assignment SDET_1.docx");  
    System.out.println("Single File Uploaded Successfully");  
    Thread.sleep(3000);  
  
    WebElement multiplefiles = driver.findElement(By.id("multipleFilesInput"));  
    multiplefiles.sendKeys("D:\\Wipro Training\\Assignment SDET_1.docx"  
        + "\n" + "D:\\Wipro Training\\Assignment SDET_2.docx");  
    System.out.println("Multiple Files Uploaded Successfully");  
    Thread.sleep(3000);  
}
```

```
public void page_table() throws InterruptedException {  
    js = (JavascriptExecutor) driver;  
    js.executeScript("window.scrollTo(0,2700)");
```

```
Thread.sleep(3000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[1]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[2]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[3]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[4]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[5]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.linkText("2")).click();
```

```
Thread.sleep(2000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[1]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[2]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[3]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[4]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[5]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.linkText("3")).click();
```

```
Thread.sleep(2000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[1]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[2]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[3]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[4]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[5]/td[4]/input")).click();
```

```
Thread.sleep(1000);
```

```
driver.findElement(By.linkText("4")).click();
```

```

Thread.sleep(2000);
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[1]/td[4]/input")).click();
Thread.sleep(1000);
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[2]/td[4]/input")).click();
Thread.sleep(1000);
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[3]/td[4]/input")).click();
Thread.sleep(1000);
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[4]/td[4]/input")).click();
Thread.sleep(1000);
driver.findElement(By.xpath("//*[@id='productTable']/tbody/tr[5]/td[4]/input")).click();
Thread.sleep(1000);

System.out.println("20 checkboxes clicked successfully in 4 pages ");
}

public void forms(String firstparagraph, String secondparagraph, String thirdparagraph)
    throws InterruptedException {
    js = (JavascriptExecutor) driver;
    js.executeScript("window.scrollTo(0,3000)");
    Thread.sleep(3000);

    driver.findElement(By.id("input1")).sendKeys(firstparagraph);
    Thread.sleep(2000);
    driver.findElement(By.id("btn1")).click();
    System.out.println("Form 1 Submitted Successfully");
    Thread.sleep(2000);

    driver.findElement(By.id("input2")).sendKeys(secondparagraph);
    Thread.sleep(2000);
    driver.findElement(By.id("btn2")).click();
    System.out.println("Form 2 Submitted Successfully");
    Thread.sleep(2000);

    driver.findElement(By.id("input3")).sendKeys(thirdparagraph);
    Thread.sleep(2000);
    driver.findElement(By.id("btn3")).click();
    System.out.println("Form 3 Submitted Successfully");
    Thread.sleep(2000);
}

```

```

public void dynamicbtn() throws InterruptedException {
    js = (JavascriptExecutor) driver;
    js.executeScript("window.scrollTo(0,900)");
    Thread.sleep(2000);

    driver.findElement(By.name("start")).click();
    System.out.println("Successfully Clicked the Dynamic Start Button");
    Thread.sleep(3000);

    driver.findElement(By.name("stop")).click();
    System.out.println("Successfully Clicked the Dynamic Stop Button");
    Thread.sleep(2000);
}

public void newtab() throws InterruptedException {
    js = (JavascriptExecutor) driver;
    js.executeScript("window.scrollTo(0,500)");
    Thread.sleep(2000);

    String oldwindow = driver.getWindowHandle();

    driver.findElement(By.xpath("//*[@id='HTML4']/div[1]/button")).click();
    Thread.sleep(3000);

    Set<String> allWindows = driver.getWindowHandles();

    for (String window : allWindows) {
        if (!window.equals(oldwindow)) {
            driver.switchTo().window(window);
        }
    }

    System.out.println("Switched to new tab: " + driver.getTitle());
    Thread.sleep(2000);

    driver.close();
    driver.switchTo().window(oldwindow);
    System.out.println("Successfully returned to parent window ");
    Thread.sleep(2000);
}

```

```

public void movetoelement() throws InterruptedException {
    js = (JavascriptExecutor) driver;
    js.executeScript("window.scrollTo(0,700)");
    Thread.sleep(2000);

    WebElement moveto =
driver.findElement(By.xpath("//*[@id='HTML3']/div[1]/div/button"));

    Actions a = new Actions(driver);
    a.moveToElement(moveto).perform();

    System.out.println("Mouse successfully hovered over element");
    Thread.sleep(3000);

}

}

```

/Automation Test Practice/src/test/java/Utilities/ExtentManager.java
package Utilities;

```

import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.reporter.ExtentSparkReporter;

public class ExtentManager {
    private static ExtentReports extent;

    public static ExtentReports getInstance() {
        if (extent == null) {
            ExtentSparkReporter spark =
                new ExtentSparkReporter("Reports/ExtentReport.html");
            extent = new ExtentReports();
            extent.attachReporter(spark);
            extent.setSystemInfo("Tester", "Sai Teja");
            extent.setSystemInfo("Project", "Automation Test Practice");
        }
        return extent;
    }
}

```

/Automation Test Practice/src/test/java/Utilities/TestListener.java

```
package Utilities;
import org.testng.*;
import com.aventstack.extentreports.*;

public class TestListener implements ITestListener {
    ExtentReports extent = ExtentManager.getInstance();
    ExtentTest test;

    @Override
    public void onTestStart(ITestResult result) {
        test = extent.createTest(result.getName());
    }

    @Override
    public void onTestSuccess(ITestResult result) {
        test.pass("Test Passed Successfully");
    }

    @Override
    public void onTestFailure(ITestResult result) {
        test.fail("Test Failed");
        test.fail(result.getThrowable());
    }

    @Override
    public void onTestSkipped(ITestResult result) {
        test.skip("Test Skipped");
    }

    @Override
    public void onFinish(ITestContext context) {
        extent.flush();
    }
}
```

/Automation Test Practice/pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>Capstone_Project</groupId>

<artifactId>Automation_Test_Practice</artifactId>

<version>0.0.1-SNAPSHOT</version>

<dependencies>

<!-- Selenium WebDriver -->

<dependency>

<groupId>org.seleniumhq.selenium</groupId>

<artifactId>selenium-java</artifactId>

<version>4.33.0</version>

</dependency>

<!-- ExtentReports for HTML Reports -->

<dependency>

<groupId>com.aventstack</groupId>

<artifactId>extentreports</artifactId>

<version>5.1.2</version>

</dependency>

<!-- TestNG Testing Framework -->

<dependency>

<groupId>org.testng</groupId>

<artifactId>testng</artifactId>

<version>7.12.0</version>

<scope>test</scope>

</dependency>

<!-- Google Guava Utilities -->

<dependency>

<groupId>com.google.guava</groupId>

<artifactId>guava</artifactId>

<version>33.2.1-jre</version>

</dependency>

<!-- Apache POI for Excel Reading -->

<dependency>

<groupId>org.apache.poi</groupId>

<artifactId>poi-ooxml</artifactId>

<version>5.4.1</version>

</dependency>

</dependencies>

<build>

```

<plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-plugin</artifactId>
    <version>3.5.3</version>
    <configuration>
      <suiteXmlFiles>
        <suiteXmlFile>testng.xml</suiteXmlFile>
      </suiteXmlFiles>
    </configuration>
  </plugin>
</plugins>
</build>
</project>

```

/Automation Test Practice/testng.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">

<suite name="AutomationSuite">

  <!-- Attach Listeners for Reporting and Logging -->
  <listeners>
    <listener class-name="Listener.Listenerclass"/>
    <listener class-name="Utilities.TestListener"/>
  </listeners>

  <!-- Define Test -->
  <test name="AllTests">
    <classes>
      <class name="Capstone_Project.BaseTest"/>
    </classes>
  </test>

</suite>

```

/Automation Test Practice/JenkinsFile

```

pipeline {
  agent any
  tools {

```

```

    maven 'Maven3'
  }
  stages {
    stage('Checkout') {
      steps {
        git(branch: 'main',
            url: 'https://github.com/Saiteja-231/Capstone-Project.git')
      }
    }
    stage('Build') {
      steps {
        dir('Automation_Test_Practice') {
          bat 'mvn clean compile'
        }
      }
    }
    stage('Test') {
      steps {
        dir('Automation_Test_Practice') {
          bat 'mvn test'
        }
      }
    }
  }
  post {
    always {
      dir('Automation_Test_Practice') {
        archiveArtifacts artifacts: 'test-output/**, Reports/**, Screenshot/**',
            allowEmptyArchive: true
      }
    }
  }
}

```

Console Output

```

=====
AutomationSuite
Total tests run: 14, Passes: 14, Failures: 0, Skips: 0
=====

```

Output Screens

